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Lab Practical #01:

Study of basic networking commands and IP configuration.

Practical Assignment #01:

1. Perform and explain various networking commands listed below:
   i. ipconfig
   ii. ping
   iii. getmac
   iv. systeminfo
   v. traceroute / tracert
   vi. netstat
   vii. nslookup
   viii. hostname
   ix. pathping
   x. arp

1. ipconfig

Description: Displays all current TCP/IP network configuration values and refreshes Dynamic Host Configuration Protocol (DHCP) and Domain Name System (DNS) settings.

No.	Option	Description
1	/all	Display full configuration information.
2	/release	Release the IPv4 address for the specified adapter.
3	/renew	Renew the IPv4 address for the specified adapter.
4	/displaydns	Display the contents of the DNS Resolver Cache.
5	/flushdns	Purges the DNS Resolver Cache.

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Implementation:

```
E:\Computer Network>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 9:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 10:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::6175:b7c2:37fb:9ae%4
   IPv4 Address. . . . . . . . . . . : 10.25.61.110
   Subnet Mask . . . . . . . . . . . : 255.255.0.0
   Default Gateway . . . . . . . . . : 10.25.1.1
```

```
E:\Computer Network>ipconfig/release

Windows IP Configuration

No operation can be performed on Ethernet 2 while it has its media disconnected.
No operation can be performed on Local Area Connection* 9 while it has its media disconnected.
No operation can be performed on Local Area Connection* 10 while it has its media disconnected.

Ethernet adapter Ethernet 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 9:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 10:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::6175:b7c2:37fb:9ae%4
   Default Gateway . . . . . . . . . :
```

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```
E:\Computer Network>ipconfig/all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : DESKTOP-66JFJH1
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Ethernet adapter Ethernet 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Realtek PCIe GbE Family Controller
   Physical Address. . . . . . . . . : B4-B6-86-D8-6C-4C
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 9:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
   Physical Address. . . . . . . . . : 82-C5-F2-FC-E5-4F
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 10:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
   Physical Address. . . . . . . . . : 80-C5-F2-FC-E5-4F
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Realtek RTL8723DE 802.11b/g/n PCIe Adapter
   Physical Address. . . . . . . . . : 80-C5-F2-FC-E5-4F
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::6175:b7c2:37fb:9ae%4(Preferred)
   IPv4 Address. . . . . . . . . . . : 10.25.61.110(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.0.0
   Lease Obtained. . . . . . . . . . : 11 June 2026 10:14:58 AM
   Lease Expires . . . . . . . . . . : 12 June 2026 10:14:58 AM
   Default Gateway . . . . . . . . . : 10.25.1.1
   DHCP Server . . . . . . . . . . . : 10.25.1.1
   DHCPv6 IAID . . . . . . . . . . . : 427869682
```

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```
E:\Computer Network>ipconfig/displaydns

Windows IP Configuration

    crs.cr.adobe.com
    ----------------------------------------
    Record Name . . . . . : crs.cr.adobe.com
    Record Type . . . . . : 1
    Time To Live  . . . . : 1883
    Data Length . . . . . : 4
    Section . . . . . . . : Answer
    A (Host) Record . . . : 192.147.130.166
```

```
E:\Computer Network>ipconfig/renew

Windows IP Configuration

No operation can be performed on Ethernet 2 while it has its media disconnected.
No operation can be performed on Local Area Connection* 9 while it has its media disconnected.
No operation can be performed on Local Area Connection* 10 while it has its media disconnected.

Ethernet adapter Ethernet 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 9:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 10:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::6175:b7c2:37fb:9ae%4
   IPv4 Address. . . . . . . . . . . : 10.25.61.110
   Subnet Mask . . . . . . . . . . . : 255.255.0.0
   Default Gateway . . . . . . . . . : 10.25.1.1
```

2. ping

Description: Sends test packets to a specific network address to verify connectivity and measure the response time.

No.	Option	Description
1	-a	Resolve address to hostname.
2	-n count	Number of echo request to send.
3	-l size	Send buffer size.
4	-i TTL	Time To Live.
5	-4	Force using IPv4.

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Implementation:

```
E:\Computer Network>ping

Usage: ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL] [-v TOS]
            [-r count] [-s count] [[-j host-list] | [-k host-list]]
            [-w timeout] [-R] [-S srcaddr] [-c compartment] [-p]
            [-4] [-6] target_name

Options:
    -t             Ping the specified host until stopped.
                   To see statistics and continue - type Control-Break;
                   To stop - type Control-C.
    -a             Resolve addresses to hostnames.
    -n count       Number of echo requests to send.
    -l size        Send buffer size.
    -f             Set Don't Fragment flag in packet (IPv4-only).
    -i TTL         Time To Live.
    -v TOS         Type Of Service (IPv4-only. This setting has been deprecated
                   and has no effect on the type of service field in the IP
                   Header).
    -r count       Record route for count hops (IPv4-only).
    -s count       Timestamp for count hops (IPv4-only).
    -j host-list   Loose source route along host-list (IPv4-only).
    -k host-list   Strict source route along host-list (IPv4-only).
    -w timeout     Timeout in milliseconds to wait for each reply.
    -R             Use routing header to test reverse route also (IPv6-only).
                   Per RFC 5095 the use of this routing header has been
                   deprecated. Some systems may drop echo requests if
                   this header is used.
    -S srcaddr     Source address to use.
    -c compartment Routing compartment identifier.
    -p             Ping a Hyper-V Network Virtualization provider address.
    -4             Force using IPv4.
    -6             Force using IPv6.
```

```
E:\Computer Network>ping -t google.com -n 5

Pinging google.com [142.250.143.100] with 32 bytes of data:
Reply from 142.250.143.100: bytes=32 time=85ms TTL=113
Reply from 142.250.143.100: bytes=32 time=108ms TTL=113
Reply from 142.250.143.100: bytes=32 time=149ms TTL=113
Reply from 142.250.143.100: bytes=32 time=96ms TTL=113
Reply from 142.250.143.100: bytes=32 time=25ms TTL=113

Ping statistics for 142.250.143.100:
    Packets: Sent = 5, Received = 5, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 25ms, Maximum = 149ms, Average = 92ms
```

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```
E:\Computer Network>ping -a 8.8.8.8 -n 5

Pinging dns.google [8.8.8.8] with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=77ms TTL=118
Reply from 8.8.8.8: bytes=32 time=55ms TTL=118
Reply from 8.8.8.8: bytes=32 time=35ms TTL=118
Reply from 8.8.8.8: bytes=32 time=33ms TTL=118
Reply from 8.8.8.8: bytes=32 time=27ms TTL=118

Ping statistics for 8.8.8.8:
    Packets: Sent = 5, Received = 5, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 27ms, Maximum = 77ms, Average = 45ms
```

```
E:\Computer Network>ping -f 127.0.0.1 -n 5

Pinging 127.0.0.1 with 32 bytes of data:
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128

Ping statistics for 127.0.0.1:
    Packets: Sent = 5, Received = 5, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
E:\Computer Network>ping -a google.com -n 5

Pinging google.com [142.250.143.102] with 32 bytes of data:
Reply from 142.250.143.102: bytes=32 time=64ms TTL=114
Reply from 142.250.143.102: bytes=32 time=49ms TTL=114
Reply from 142.250.143.102: bytes=32 time=30ms TTL=114
Reply from 142.250.143.102: bytes=32 time=22ms TTL=114
Reply from 142.250.143.102: bytes=32 time=83ms TTL=114

Ping statistics for 142.250.143.102:
    Packets: Sent = 5, Received = 5, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 22ms, Maximum = 83ms, Average = 49ms
```

```
E:\Computer Network>ping -t 8.8.8.8 -n 5

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=92ms TTL=118
Reply from 8.8.8.8: bytes=32 time=93ms TTL=118
Reply from 8.8.8.8: bytes=32 time=84ms TTL=118
Reply from 8.8.8.8: bytes=32 time=80ms TTL=118
Reply from 8.8.8.8: bytes=32 time=77ms TTL=118

Ping statistics for 8.8.8.8:
    Packets: Sent = 5, Received = 5, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 77ms, Maximum = 93ms, Average = 85ms
```

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3. getmac

Description: Returns the Media Access Control (MAC) address and list of network protocols associated with each network adapter in the computer.

No.	Option	Description
1	/S system	Specifies the remote system to connect to.
2	/U user	Specifies the user context under which the command should execute.
3	/P password	Specifies the password for the given use context. Prompts for input if omitted.
4	/FO format	Specifies the format in which the output is to be displayed.
5	/NH	Specifies that the "Column Header" should not be displayed in the output. Vaild only for TABLE and CSV formats.

Implementation:

```
E:\Computer Network>getmac

Physical Address    Transport Name
=================== ============================================================
B4-B6-86-D8-6C-4C   Media disconnected
80-C5-F2-FC-E5-4F   \Device\Tcpip_{0B08C9AA-91DA-4CA1-9498-4AD23F62A8A6}
```

```
E:\Computer Network>getmac /v

Connection Name Network Adapter Physical Address    Transport Name
=============== =============== =================== ============================================================
Ethernet 2      Realtek PCIe Gb B4-B6-86-D8-6C-4C   Media disconnected
Wi-Fi           Realtek RTL8723 80-C5-F2-FC-E5-4F   \Device\Tcpip_{0B08C9AA-91DA-4CA1-9498-4AD23F62A8A6}
```

```
E:\Computer Network>getmac /fo csv
"Physical Address","Transport Name"
"B4-B6-86-D8-6C-4C","Media disconnected"
"80-C5-F2-FC-E5-4F","\Device\Tcpip_{0B08C9AA-91DA-4CA1-9498-4AD23F62A8A6}"
```

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```
E:\Computer Network>getmac/?

GETMAC [/S system [/U username [/P [password]]]] [/FO format] [/NH] [/V]

Description:
    This tool enables an administrator to display the MAC address
    for network adapters on a system.

Parameter List:
    /S     system           Specifies the remote system to connect to.

    /U     [domain\]user    Specifies the user context under
                            which the command should execute.

    /P     [password]       Specifies the password for the given
                            user context. Prompts for input if omitted.

    /FO    format           Specifies the format in which the output
                            is to be displayed.
                            Valid values: "TABLE", "LIST", "CSV".

    /NH                     Specifies that the "Column Header" should
                            not be displayed in the output.
                            Valid only for TABLE and CSV formats.

    /V                      Specifies that verbose output is displayed.

    /?                      Displays this help message.

Examples:
    GETMAC /?
    GETMAC /FO csv
    GETMAC /S system /NH /V
    GETMAC /S system /U user
    GETMAC /S system /U domain\user /P password /FO list /V
    GETMAC /S system /U domain\user /P password /FO table /NH
```

```
E:\Computer Network>getmac /s DESKTOP-66JFJH1 /u DESKTOP-66JFJH1\user /p ronit@123
WARNING: User credentials cannot be used for local connections.

Physical Address    Transport Name
=================== ============================================================
B4-B6-86-D8-6C-4C   Media disconnected
80-C5-F2-FC-E5-4F   \Device\Tcpip_{0B08C9AA-91DA-4CA1-9498-4AD23F62A8A6}
```

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4. systeminfo

Description: Displays detailed configuration information about your computer and its operating system, including hardware properties and system updates.

No.	Option	Description
1	/S system	Specifies the remote system to connect to.
2	/U user	Specifies the user context under which the command should execute.
3	/P password	Specifies the password for the given use context. Prompts for input if omitted.
4	/FO format	Specifies the format in which the output is to be displayed.
5	/NH	Specifies that the "Column Header" should not be displayed in the output. Vaild only for TABLE and CSV formats.

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Implementation:

```
E:\Computer Network>systeminfo

Host Name:                 DESKTOP-66JFJH1
OS Name:                   Microsoft Windows 10 Home Single Language
OS Version:                10.0.19045 N/A Build 19045
OS Manufacturer:           Microsoft Corporation
OS Configuration:          Standalone Workstation
OS Build Type:             Multiprocessor Free
Registered Owner:          User
Registered Organization:
Product ID:                00327-35072-31431-AAOEM
Original Install Date:     20-05-2023, 12:06:55 AM
System Boot Time:          11-06-2026, 09:49:02 AM
System Manufacturer:       HP
System Model:              HP Laptop 14q-bu0xx
System Type:               x64-based PC
Processor(s):              1 Processor(s) Installed.
                           [01]: Intel64 Family 6 Model 78 Stepping 3 GenuineIntel ~1992 Mhz
BIOS Version:              Insyde F.32, 26-02-2018
Windows Directory:         C:\Windows
System Directory:          C:\Windows\system32
Boot Device:               \Device\HarddiskVolume3
System Locale:             en-us;English (United States)
Input Locale:              en-us;English (United States)
Time Zone:                 (UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi
Total Physical Memory:     12,174 MB
Available Physical Memory: 6,928 MB
Virtual Memory: Max Size:  14,030 MB
Virtual Memory: Available: 8,064 MB
Virtual Memory: In Use:    5,966 MB
Page File Location(s):     C:\pagefile.sys
Domain:                    WORKGROUP
Logon Server:              \\DESKTOP-66JFJH1
Hotfix(s):                 26 Hotfix(s) Installed.
                           [01]: KB5066130
                           [02]: KB5042056
                           [03]: KB5011048
                           [04]: KB5015684
                           [05]: KB5033052
                           [06]: KB5072653
                           [07]: KB5071959
                           [08]: KB5014032
                           [09]: KB5025315
```

```
E:\Computer Network>systeminfo | findstr /B /C:"OS Name" /C:"OS Version"
OS Name:                   Microsoft Windows 10 Home Single Language
OS Version:                10.0.19045 N/A Build 19045
```

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```
E:\Computer Network>systeminfo | findstr /B /C:"System Type" /C:"Total Physical Memory"
System Type:               x64-based PC
Total Physical Memory:     12,174 MB
```

```
E:\Computer Network>systeminfo | findstr /C:"Network Card"
Network Card(s):           2 NIC(s) Installed.
```

```
E:\Computer Network>systeminfo /?

SYSTEMINFO [/S system [/U username [/P [password]]]] [/FO format] [/NH]

Description:
    This tool displays operating system configuration information for
    a local or remote machine, including service pack levels.

Parameter List:
    /S      system          Specifies the remote system to connect to.

    /U      [domain\]user   Specifies the user context under which
                            the command should execute.

    /P      [password]      Specifies the password for the given
                            user context. Prompts for input if omitted

    /FO     format          Specifies the format in which the output
                            is to be displayed.
                            Valid values: "TABLE", "LIST", "CSV".

    /NH                     Specifies that the "Column Header" should
                            not be displayed in the output.
                            Valid only for "TABLE" and "CSV" formats.

    /?                      Displays this help message.

Examples:
    SYSTEMINFO
    SYSTEMINFO /?
    SYSTEMINFO /S system
    SYSTEMINFO /S system /U user
    SYSTEMINFO /S system /U domain\user /P password /FO TABLE
    SYSTEMINFO /S system /FO LIST
    SYSTEMINFO /S system /FO CSV /NH
```

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5. tracert

Description: Tracks and displays the exact path (and each router hop) that a packet takes to reach a specific destination.

No.	Option	Description
1	-d	Do not resolve address to hostnames.
2	-h maximum_hops	Maximum number of hops to search for target.
3	-j host-list	Loose source route along host-list.(IPv4-only)
4	-w timeout	Wait timeout milliseconds for each reply.
5	-R	Trace round-trip path.(IPv6 only)

Implementation:

```
E:\Computer Network>tracert

Usage: tracert [-d] [-h maximum_hops] [-j host-list] [-w timeout]
               [-R] [-S srcaddr] [-4] [-6] target_name

Options:
    -d                 Do not resolve addresses to hostnames.
    -h maximum_hops    Maximum number of hops to search for target.
    -j host-list       Loose source route along host-list (IPv4-only).
    -w timeout         Wait timeout milliseconds for each reply.
    -R                 Trace round-trip path (IPv6-only).
    -S srcaddr         Source address to use (IPv6-only).
    -4                 Force using IPv4.
    -6                 Force using IPv6.
```

```
E:\Computer Network>tracert -h 6 google.com

Tracing route to google.com [142.250.134.100]
over a maximum of 6 hops:

  1    61 ms     2 ms     3 ms  10.120.1.1
  2   151 ms    96 ms     5 ms  static.ill.117.250.171.130.bsnl.co.in [117.250.171.130]
  3     *         *        *     Request timed out.
  4     *         *        *     Request timed out.
  5     *         *        *     Request timed out.
  6     *         *        *     Request timed out.

Trace complete.
```

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```
E:\Computer Network>tracert -4 google.com

Tracing route to google.com [142.250.143.100]
over a maximum of 30 hops:

  1    63 ms     2 ms     7 ms  10.120.1.1
  2    51 ms    95 ms    95 ms  static.ill.117.250.171.130.bsnl.co.in [117.250.171.130]
  3     *        *        *     Request timed out.
  4     *        *        *     Request timed out.
  5     *        *        *     Request timed out.
  6     *        *        *     Request timed out.
  7     *        *        *     Request timed out.
  8    54 ms    98 ms    97 ms  72.14.197.4
  9    63 ms    23 ms    21 ms  142.251.225.9
 10   124 ms    96 ms    96 ms  172.253.177.30
 11   163 ms    97 ms   100 ms  142.251.230.117
 12    59 ms    99 ms    19 ms  209.85.246.70
 13     *        *        *     Request timed out.
 14     *        *        *     Request timed out.
 15     *        *        *     Request timed out.
 16     *        *        *     Request timed out.
 17     *        *        *     Request timed out.
 18     *        *        *     Request timed out.
 19   133 ms    97 ms   101 ms  pt-in-f100.1e100.net [142.250.143.100]

Trace complete.
```

```
E:\Computer Network>tracert -d github.com

Tracing route to github.com [20.207.73.82]
over a maximum of 30 hops:

  1    65 ms     2 ms     2 ms  10.120.1.1
  2     6 ms     7 ms     7 ms  117.250.171.130
  3     *        *        *     Request timed out.
  4     *        *        *     Request timed out.
  5     *        *        *     Request timed out.
  6     *        *        *     Request timed out.
  7     *        *        *     Request timed out.
  8     *        *        *     Request timed out.
  9     *       18 ms    20 ms  104.44.230.0
 10     *        *        *     Request timed out.
 11     *        *        *     Request timed out.
 12     *        *        *     Request timed out.
 13    97 ms     *       20 ms  104.44.20.46
 14     *        *        *     Request timed out.
 15     *        *        *     Request timed out.
 16     *        *        *     Request timed out.
 17     *        *        *     Request timed out.
 18    55 ms   196 ms    94 ms  20.207.73.82

Trace complete.
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```
E:\Computer Network>tracert -w 5 google.com

Tracing route to google.com [142.251.220.14]
over a maximum of 30 hops:

  1     2 ms     1 ms     1 ms  10.120.1.1
  2     3 ms     3 ms     8 ms  static.ill.117.250.171.130.bsnl.co.in [117.250.171.130]
  3     *        *        *     Request timed out.
  4     *        *        *     Request timed out.
  5     *        *        *     Request timed out.
  6    18 ms    34 ms    27 ms  10.200.45.62
  7     *        *        *     Request timed out.
  8    17 ms    20 ms    25 ms  72.14.197.4
  9    24 ms    14 ms    14 ms  216.239.47.175
 10    15 ms    18 ms    17 ms  142.251.64.13
 11    15 ms    16 ms    17 ms  pnbomb-ay-in-f14.1e100.net [142.251.220.14]

Trace complete.
```

6. netstat

Description: Displays active network connections, listening ports, and routing tables to help monitor network traffic and security.

No.	Option	Description
1	-a	Displays all connections and listening ports.
2	-c	Displays a list of processes sorted by the number of TCP or UDP ports currently consumed.
3	-d	Displays DSCP value assocated with each connection
4	-e	Displays Ethernet statistics. This may be combined with the -s option.
5	-f	Displays Fully Qualified Domain Names(FQDN) for foreign addresses.

Implementation:

```
E:\Computer Network>netstat -e
Interface Statistics

                           Received            Sent

Bytes                    1637536476      1207988064
Unicast packets             1259208          704766
Non-unicast packets          485034            1938
Discards                          0               0
Errors                            0               0
Unknown protocols                 0
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```
E:\Computer Network>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    10.120.15.176:50409    relay-69864d51:https   ESTABLISHED
  TCP    10.120.15.176:50415    4.213.25.240:https     ESTABLISHED
  TCP    10.120.15.176:50427    a23-218-249-184:https  CLOSE_WAIT
  TCP    10.120.15.176:50435    52.98.34.194:https     ESTABLISHED
  TCP    10.120.15.176:50457    4.213.25.241:https     ESTABLISHED
  TCP    10.120.15.176:50485    0:https                ESTABLISHED
  TCP    10.120.15.176:50516    104.18.39.21:https     ESTABLISHED
  TCP    10.120.15.176:50518    si-in-f188:5228        ESTABLISHED
  TCP    10.120.15.176:50643    40.104.66.162:https    ESTABLISHED
  TCP    10.120.15.176:50644    40.104.66.162:https    ESTABLISHED
  TCP    10.120.15.176:50648    a184-51-94-9:https     ESTABLISHED
  TCP    10.120.15.176:50651    lcboma-be-in-f10:https  TIME_WAIT
  TCP    10.120.15.176:50666    server-13-227-192-56:https  ESTABLISHED
  TCP    10.120.15.176:50668    lb-140-82-114-22-iad:https  ESTABLISHED
  TCP    10.120.15.176:50671    13.107.5.93:https      ESTABLISHED
  TCP    10.120.15.176:50672    20.207.73.85:https     ESTABLISHED
  TCP    127.0.0.1:49673        DESKTOP-66JFJH1:49674  ESTABLISHED
  TCP    127.0.0.1:49674        DESKTOP-66JFJH1:49673  ESTABLISHED
  TCP    127.0.0.1:49675        DESKTOP-66JFJH1:49676  ESTABLISHED
  TCP    127.0.0.1:49676        DESKTOP-66JFJH1:49675  ESTABLISHED
```

```
E:\Computer Network>netstat -o

Active Connections

  Proto  Local Address          Foreign Address        State           PID
  TCP    10.120.15.176:50409    relay-69864d51:https   ESTABLISHED     3392
  TCP    10.120.15.176:50415    4.213.25.240:https     ESTABLISHED     3924
  TCP    10.120.15.176:50427    a23-218-249-184:https  CLOSE_WAIT      7096
  TCP    10.120.15.176:50435    52.98.34.194:https     ESTABLISHED     7096
  TCP    10.120.15.176:50457    4.213.25.241:https     ESTABLISHED     3964
  TCP    10.120.15.176:50516    104.18.39.21:https     ESTABLISHED     7628
  TCP    10.120.15.176:50518    si-in-f188:5228        ESTABLISHED     4624
  TCP    10.120.15.176:50643    40.104.66.162:https    ESTABLISHED     632
  TCP    10.120.15.176:50644    40.104.66.162:https    ESTABLISHED     632
  TCP    10.120.15.176:50679    lb-140-82-113-21-iad:https  ESTABLISHED     4624
  TCP    10.120.15.176:50694    server-13-227-192-46:https  ESTABLISHED     4624
  TCP    10.120.15.176:50698    98.70.192.224:https    ESTABLISHED     3084
  TCP    10.120.15.176:50699    20.207.73.85:https     ESTABLISHED     4624
  TCP    127.0.0.1:49673        DESKTOP-66JFJH1:49674  ESTABLISHED     4440
  TCP    127.0.0.1:49674        DESKTOP-66JFJH1:49673  ESTABLISHED     4440
  TCP    127.0.0.1:49675        DESKTOP-66JFJH1:49676  ESTABLISHED     4440
  TCP    127.0.0.1:49676        DESKTOP-66JFJH1:49675  ESTABLISHED     4440
```

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```
E:\Computer Network>netstat -r
===========================================================================
Interface List
 16...b4 b6 86 d8 6c 4c ......Realtek PCIe GbE Family Controller
 15...82 c5 f2 fc e5 4f ......Microsoft Wi-Fi Direct Virtual Adapter
 13...80 c5 f2 fc e5 4f ......Microsoft Wi-Fi Direct Virtual Adapter #2
  4...80 c5 f2 fc e5 4f ......Realtek RTL8723DE 802.11b/g/n PCIe Adapter
  1...........................Software Loopback Interface 1
===========================================================================

IPv4 Route Table
===========================================================================
Active Routes:
Network Destination        Netmask          Gateway       Interface  Metric
          0.0.0.0          0.0.0.0      10.120.1.1    10.120.15.176     55
       10.120.0.0      255.255.0.0         On-link     10.120.15.176    311
    10.120.15.176  255.255.255.255         On-link     10.120.15.176    311
   10.120.255.255  255.255.255.255         On-link     10.120.15.176    311
        127.0.0.0        255.0.0.0         On-link         127.0.0.1    331
        127.0.0.1  255.255.255.255         On-link         127.0.0.1    331
  127.255.255.255  255.255.255.255         On-link         127.0.0.1    331
        224.0.0.0        240.0.0.0         On-link         127.0.0.1    331
        224.0.0.0        240.0.0.0         On-link     10.120.15.176    311
  255.255.255.255  255.255.255.255         On-link         127.0.0.1    331
  255.255.255.255  255.255.255.255         On-link     10.120.15.176    311
===========================================================================
Persistent Routes:
  None

IPv6 Route Table
===========================================================================
Active Routes:
 If Metric Network Destination      Gateway
  1    331 ::1/128                  On-link
  4    311 fe80::/64                On-link
  4    311 fe80::6175:b7c2:37fb:9ae/128
                                    On-link
  1    331 ff00::/8                 On-link
  4    311 ff00::/8                 On-link
===========================================================================
Persistent Routes:
  None
```

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```
E:\Computer Network>netstat -f

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    10.120.15.176:7680     10.120.15.137:63661    ESTABLISHED
  TCP    10.120.15.176:7680     DESKTOP-46ORHHG:58484  TIME_WAIT
  TCP    10.120.15.176:7680     DESKTOP-46ORHHG:60170  TIME_WAIT
  TCP    10.120.15.176:49676    relay-8f76c373.net.anydesk.com:https  ESTABLISHED
  TCP    10.120.15.176:49678    4.213.25.241:https     ESTABLISHED
  TCP    10.120.15.176:49739    4.213.25.242:https     ESTABLISHED
  TCP    10.120.15.176:49758    172.64.148.235:https   ESTABLISHED
  TCP    10.120.15.176:49784    52.98.34.194:https     ESTABLISHED
  TCP    10.120.15.176:49786    whatsapp-cdn-shv-02-bom5.fbcdn.net:https  ESTABLISHED
  TCP    10.120.15.176:49834    ec2-34-238-100-42.compute-1.amazonaws.com:https  ESTABLISHED
  TCP    10.120.15.176:49835    ec2-57-180-122-71.ap-northeast-1.compute.amazonaws.com:https  ESTABLISHED
  TCP    10.120.15.176:49919    52.98.59.18:https      ESTABLISHED
  TCP    10.120.15.176:49920    52.98.88.66:https      ESTABLISHED
  TCP    10.120.15.176:50016    4.188.92.87:https      ESTABLISHED
  TCP    10.120.15.176:50018    ec2-52-73-45-44.compute-1.amazonaws.com:https  ESTABLISHED
  TCP    10.120.15.176:50049    150.171.27.12:https    ESTABLISHED
  TCP    10.120.15.176:50050    a104-95-190-56.deploy.static.akamaitechnologies.com:https  CLOSE_WAIT
  TCP    10.120.15.176:50052    20.184.175.19:https    ESTABLISHED
  TCP    10.120.15.176:50053    20.190.146.34:https    ESTABLISHED
  TCP    10.120.15.176:50054    20.99.186.246:https    ESTABLISHED
  TCP    10.120.15.176:50055    20.89.194.154:https    ESTABLISHED
  TCP    127.0.0.1:49672        DESKTOP-66JFJH1:49673  ESTABLISHED
  TCP    127.0.0.1:49673        DESKTOP-66JFJH1:49672  ESTABLISHED
  TCP    127.0.0.1:49674        DESKTOP-66JFJH1:49675  ESTABLISHED
  TCP    127.0.0.1:49675        DESKTOP-66JFJH1:49674  ESTABLISHED
```

7. nslookup

Description: Queries Domain Name System (DNS) servers to find the IP address associated with a domain name, or vice versa.

No.	Option	Description
1	nslookup [-opt…]	# interactive mode using default server
2	nslookup [-opt…] -server	# interactive mode using server
3	nslookup [-opt…] host	# just look up 'host' using default server
4	nslookup [-opt…] host server	# just look up 'host' using server

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Implementation:

```
E:\Computer Network>nslookup
Default Server:  dns.google
Address:  8.8.8.8

> set type=A
> google.com
Server:  dns.google
Address:  8.8.8.8

Non-authoritative answer:
Name:    google.com
Addresses:  192.178.193.101
          192.178.193.113
          192.178.193.102
          192.178.193.100
          192.178.193.139
          192.178.193.138
```

```
E:\Computer Network>nslookup google.com
Server:  UnKnown
Address:  10.25.1.1

Non-authoritative answer:
Name:    google.com
Addresses:  2404:6800:4000:100c::71
          2404:6800:4000:100c::64
          2404:6800:4000:100c::66
          2404:6800:4000:100c::8a
          142.250.143.101
          142.250.143.139
          142.250.143.102
          142.250.143.138
          142.250.143.100
          142.250.143.113
```

```
E:\Computer Network>nslookup darshan.ac.in
Server:  UnKnown
Address:  10.25.1.1

Non-authoritative answer:
Name:    darshan.ac.in
Address:  3.6.31.165
```

```
E:\Computer Network>nslookup /?
Usage:
   nslookup [-opt ...]             # interactive mode using default server
   nslookup [-opt ...] - server    # interactive mode using 'server'
   nslookup [-opt ...] host        # just look up 'host' using default server
   nslookup [-opt ...] host server # just look up 'host' using 'server'
```

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```
E:\Computer Network>nslookup
Default Server:  dns.google
Address:  8.8.8.8

> set type=ns
> darshan.ac.in
Server:  dns.google
Address:  8.8.8.8

Non-authoritative answer:
darshan.ac.in   nameserver = naomi.ns.cloudflare.com
darshan.ac.in   nameserver = ivan.ns.cloudflare.com
```

8. hostname

Description: Displays the current, user-defined name of your computer or device on the network.

No.	Option	Description
1	hostname	Display hostname of system.

Implementation:

```
E:\Computer Network>hostname
DESKTOP-66JFJH1
```

```
E:\Computer Network>echo %COMPUTERNAME%
DESKTOP-66JFJH1
```

9. pathping

Description: Combines the features of ping and tracert by tracing the route to a destination and calculating packet loss at each hop to find network bottlenecks.

No.	Option	Description
1	-g host-list	Loose soource route along host-list.
2	-h maximum_hops	Maximum number of hops to search for target.
3	-i address	Use the specified source address.
4	-n	Do not resolve addresses to hostnames.
5	-p period	Wait period miliseconds betwwen pings.

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Implementation:

```
E:\Computer Network>pathping

Usage: pathping [-g host-list] [-h maximum_hops] [-i address] [-n]
                [-p period] [-q num_queries] [-w timeout]
                [-4] [-6] target_name

Options:
    -g host-list     Loose source route along host-list.
    -h maximum_hops  Maximum number of hops to search for target.
    -i address       Use the specified source address.
    -n               Do not resolve addresses to hostnames.
    -p period        Wait period milliseconds between pings.
    -q num_queries   Number of queries per hop.
    -w timeout       Wait timeout milliseconds for each reply.
    -4               Force using IPv4.
    -6               Force using IPv6.
```

```
E:\Computer Network>pathping -h 5 github.com

Tracing route to github.com [20.207.73.82]
over a maximum of 5 hops:
  0  DESKTOP-66JFJH1 [10.120.15.176]
  1  10.120.1.1
  2  static.ill.117.250.171.130.bsnl.co.in [117.250.171.130]
  3     *        *        *
Computing statistics for 50 seconds...
            Source to Here   This Node/Link
Hop  RTT    Lost/Sent = Pct  Lost/Sent = Pct  Address
  0                                           DESKTOP-66JFJH1 [10.120.15.176]
                                    0/ 100 =  0%   |
  1    5ms     0/ 100 =  0%     0/ 100 =  0%  10.120.1.1
                                    0/ 100 =  0%   |
  2    9ms     0/ 100 =  0%     0/ 100 =  0%  static.ill.117.250.171.130.bsnl.co.in [117.250.171.130]

Trace complete.
```

```
E:\Computer Network>pathping -n google.com

Tracing route to google.com [142.250.76.206]
over a maximum of 30 hops:
  0  10.120.15.176
  1  10.120.1.1
  2  117.250.171.130
  3     *        *        *
Computing statistics for 50 seconds...
            Source to Here   This Node/Link
Hop  RTT    Lost/Sent = Pct  Lost/Sent = Pct  Address
  0                                           10.120.15.176
                                    0/ 100 =  0%   |
  1    6ms     0/ 100 =  0%     0/ 100 =  0%  10.120.1.1
                                    0/ 100 =  0%   |
  2    8ms     0/ 100 =  0%     0/ 100 =  0%  117.250.171.130

Trace complete.
```

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```
E:\Computer Network>pathping -p 10 google.com

Tracing route to google.com [142.250.134.100]
over a maximum of 30 hops:
  0  DESKTOP-66JFJH1 [10.120.15.176]
  1  10.120.1.1
  2  static.ill.117.250.171.130.bsnl.co.in [117.250.171.130]
  3     *        *        *
Computing statistics for 2 seconds...
            Source to Here   This Node/Link
Hop  RTT    Lost/Sent = Pct  Lost/Sent = Pct  Address
  0                                           DESKTOP-66JFJH1 [10.120.15.176]
                                    0/ 100 =  0%   |
  1    7ms     0/ 100 =  0%     0/ 100 =  0%  10.120.1.1
                                    0/ 100 =  0%   |
  2    9ms     0/ 100 =  0%     0/ 100 =  0%  static.ill.117.250.171.130.bsnl.co.in [117.250.171.130]

Trace complete.
```

```
E:\Computer Network>pathping -q 5 github.com

Tracing route to github.com [20.207.73.82]
over a maximum of 30 hops:
  0  DESKTOP-66JFJH1 [10.120.15.176]
  1  10.120.1.1
  2  static.ill.117.250.171.130.bsnl.co.in [117.250.171.130]
  3     *        *        *
Computing statistics for 2 seconds...
            Source to Here   This Node/Link
Hop  RTT    Lost/Sent = Pct  Lost/Sent = Pct  Address
  0                                           DESKTOP-66JFJH1 [10.120.15.176]
                                    0/   5 =  0%   |
  1   19ms     0/   5 =  0%     0/   5 =  0%  10.120.1.1
                                    0/   5 =  0%   |
  2   24ms     0/   5 =  0%     0/   5 =  0%  static.ill.117.250.171.130.bsnl.co.in [117.250.171.130]

Trace complete.
```

10. arp

Description: Displays and modifies the Address Resolution Protocol (ARP) cache, which maps local IP addresses to their physical MAC addresses.

No.	Option	Description
1	-g	Displays current ARP entries by interrogating the current protocol data.
2	Intent_addr	Specifies an internet address.
3	-N if_addr	Displays the ARP entriesfor the network interface specified by if_addr
4	-d	Deletes the host specified by intent_addr. intent_addr may be wildcarded with * delete all hosts.
5	Eth_addr	Specifies a phhysical address.

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Implementation:

```
E:\Computer Network>arp

Displays and modifies the IP-to-Physical address translation tables used by
address resolution protocol (ARP).

ARP -s inet_addr eth_addr [if_addr]
ARP -d inet_addr [if_addr]
ARP -a [inet_addr] [-N if_addr] [-v]

  -a            Displays current ARP entries by interrogating the current
                protocol data.  If inet_addr is specified, the IP and Physical
                addresses for only the specified computer are displayed.  If
                more than one network interface uses ARP, entries for each ARP
                table are displayed.
  -g            Same as -a.
  -v            Displays current ARP entries in verbose mode.  All invalid
                entries and entries on the loop-back interface will be shown.
  inet_addr     Specifies an internet address.
  -N if_addr    Displays the ARP entries for the network interface specified
                by if_addr.
  -d            Deletes the host specified by inet_addr. inet_addr may be
                wildcarded with * to delete all hosts.
  -s            Adds the host and associates the Internet address inet_addr
                with the Physical address eth_addr.  The Physical address is
                given as 6 hexadecimal bytes separated by hyphens. The entry
                is permanent.
  eth_addr      Specifies a physical address.
  if_addr       If present, this specifies the Internet address of the
                interface whose address translation table should be modified.
                If not present, the first applicable interface will be used.
Example:
  > arp -s 157.55.85.212   00-aa-00-62-c6-09  .... Adds a static entry.
  > arp -a                                    .... Displays the arp table.
```

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```
E:\Computer Network>arp -a

Interface: 10.120.15.176 --- 0x4
  Internet Address      Physical Address      Type
  10.120.1.1            7c-5a-1c-ce-2f-57     dynamic
  10.120.15.14          00-42-38-ca-8c-bc     dynamic
  10.120.15.35          d0-65-78-91-0d-6b     dynamic
  10.120.15.46          bc-03-58-77-c4-e6     dynamic
  10.120.15.53          10-68-38-82-61-c0     dynamic
  10.120.15.54          f8-54-f6-22-96-b5     dynamic
  10.120.15.56          cc-5e-f8-bd-44-0d     dynamic
  10.120.15.65          4e-0f-9b-2e-d9-bc     dynamic
  10.120.15.104         c2-4b-3d-f7-c1-96     dynamic
  10.120.15.138         28-0c-50-56-21-1a     dynamic
  10.120.15.194         68-7a-64-18-e2-b9     dynamic
  10.120.15.240         6a-f4-1b-77-d8-29     dynamic
  10.120.15.242         78-af-08-18-ac-68     dynamic
  10.120.16.3           72-70-05-49-d2-6b     dynamic
  10.120.16.14          84-9e-56-bf-c0-71     dynamic
  10.120.16.48          94-bb-43-42-2d-bf     dynamic
  10.120.16.128         c0-35-32-14-88-05     dynamic
  10.120.255.255        ff-ff-ff-ff-ff-ff     static
  224.0.0.2             01-00-5e-00-00-02     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.120           01-00-5e-00-00-78     static
  224.0.0.250           01-00-5e-00-00-fa     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  224.0.1.187           01-00-5e-00-01-bb     static
  239.2.0.252           01-00-5e-02-00-fc     static
  239.255.102.18        01-00-5e-7f-66-12     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static
```

```
E:\Computer Network>arp -g

Interface: 10.120.15.176 --- 0x4
  Internet Address      Physical Address      Type
  10.120.1.1            7c-5a-1c-ce-2f-57     dynamic
  10.120.15.14          00-42-38-ca-8c-bc     dynamic
  10.120.15.35          d0-65-78-91-0d-6b     dynamic
  10.120.15.46          bc-03-58-77-c4-e6     dynamic
  10.120.15.53          10-68-38-82-61-c0     dynamic
  10.120.15.54          f8-54-f6-22-96-b5     dynamic
  10.120.15.56          cc-5e-f8-bd-44-0d     dynamic
  10.120.15.65          4e-0f-9b-2e-d9-bc     dynamic
  10.120.15.104         c2-4b-3d-f7-c1-96     dynamic
  10.120.15.138         28-0c-50-56-21-1a     dynamic
  10.120.15.194         68-7a-64-18-e2-b9     dynamic
  10.120.15.240         6a-f4-1b-77-d8-29     dynamic
  10.120.15.242         78-af-08-18-ac-68     dynamic
  10.120.16.3           72-70-05-49-d2-6b     dynamic
  10.120.16.14          84-9e-56-bf-c0-71     dynamic
  10.120.16.48          94-bb-43-42-2d-bf     dynamic
  10.120.16.128         c0-35-32-14-88-05     dynamic
  10.120.255.255        ff-ff-ff-ff-ff-ff     static
  224.0.0.2             01-00-5e-00-00-02     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.120           01-00-5e-00-00-78     static
  224.0.0.250           01-00-5e-00-00-fa     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  224.0.1.187           01-00-5e-00-01-bb     static
  239.2.0.252           01-00-5e-02-00-fc     static
  239.255.102.18        01-00-5e-7f-66-12     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static
```

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```
E:\Computer Network>arp -a -v

Interface: 127.0.0.1 --- 0x1
  Internet Address      Physical Address      Type
  224.0.0.2                                   static
  224.0.0.22                                  static
  224.0.0.120                                 static
  224.0.0.253                                 static
  224.0.2.60                                  static
  224.77.77.77                                static
  239.2.0.252                                 static
  239.192.152.143                             static
  239.255.255.250                             static

Interface: 10.120.15.176 --- 0x4
  Internet Address      Physical Address      Type
  10.120.1.1            7c-5a-1c-ce-2f-57     dynamic
  10.120.15.14          00-42-38-ca-8c-bc     dynamic
  10.120.15.35          d0-65-78-91-0d-6b     dynamic
  10.120.15.46          bc-03-58-77-c4-e6     dynamic
  10.120.15.137         60-45-2e-6c-87-2a     dynamic
  10.120.15.146         cc-47-40-39-5b-8b     dynamic
  10.120.15.163         60-ff-9e-45-aa-68     dynamic
  10.120.15.241         e4-a4-71-3b-ba-50     dynamic
  10.120.16.12          38-d5-7a-07-7f-e3     dynamic
  10.120.16.24          60-45-2e-7e-0b-21     dynamic
  10.120.16.48          94-bb-43-42-2d-bf     dynamic
  10.120.16.108         5c-5f-67-07-7c-c7     dynamic
  10.120.16.125         a8-41-f4-b4-8b-e7     dynamic
  10.120.16.129         2e-e5-b8-7d-03-63     dynamic
  10.120.16.164         16-6f-19-a6-d3-8e     dynamic
  10.120.16.230         e4-0d-36-9f-d0-26     dynamic
  10.120.17.11          00-2d-f4-70-05-77     dynamic
  10.120.22.128         18-47-3d-e3-8a-b9     dynamic
  10.120.255.255        ff-ff-ff-ff-ff-ff     static
  169.254.169.254       00-00-00-00-00-00     invalid
  224.0.0.2             01-00-5e-00-00-02     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.120           01-00-5e-00-00-78     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  224.0.0.253           01-00-5e-00-00-fd     static
  224.0.2.60            01-00-5e-00-02-3c     static
```